CprE/SE 4910 WEEKLY REPORT 04

10/07/2026 – 10/13/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

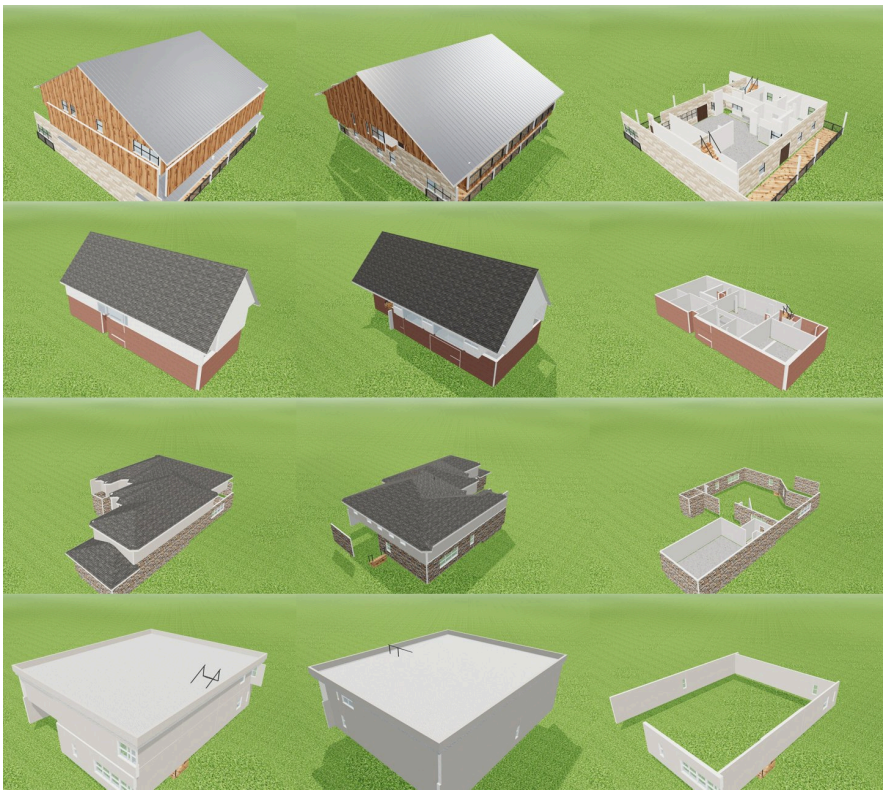
Weekly Summary

The group had the first meeting with the client this week. This meeting changed the views on what the project entitled quite a bit. This caused some members of the team to start researching how to circumvent new problems that were initially not known as well as open new work for the prototype.

Past Week Accomplishments

- Group Work:
  - Had the first meeting with the client and discussed the future of the project and what needs to be done.
- Nakshatra Gupta:
  - Analyzed one of the client's sample plans, a 21-sheet CAD-exported PDF with a readable text layer. Dimensions and door and window schedules can be read directly as text, which may reduce how much of the plan needs computer vision.
  - Wrote a Python script (PyMuPDF) that classifies sheets and extracts the schedules, dimensions, roof pitches and area table into JSON. On the sample plan it found 23 doors and 25 windows, and the tags on the floor plans matched the schedules.
  - Emailed the faculty advisor with the website link and a client meeting update.
- Rishiraj Parmar:

    - Reviewed the Buildertrend-provided two-story build plans to identify the information required for multi-story reconstruction. Focused on floor relationships, floor elevations, story heights, stairs, open-to-below areas, and roof relationships.
    - Developed an initial multi-story reconstruction integration prototype in Python that accepts already-structured building data and organizes it by floor.
    - Identified proposed story-level fields such as story ID, level number, base elevation, story height, associated walls, stairs, and vertical relationships between floors.
- Ryan McNab:
    - Improved the prototype using the sample blueprints received from Buildertrend.
    - Created about 100 new materials that come up in the sample blueprints, as well as some more common ones that might show up in the future.
    - Allows full documents as input and parses the floorplans and other information out of it, even though it is still flawed at this point in time.
    - Adds second floors when there is two floors in the blueprint.
    - Reads the roof type from the documents if specified.
    - Adds new meshes for porch fences and pillars along with balcony versions too.
    - Adds layered view of the floors, so the first floor can be viewed from orbital viewer easier.
    - Example of outputs from the sample floorplans:



- Daniel Hargrave:
    - After the meeting with the client, studied the received floor plans fairly

extensively
    - Tried to look into what is necessary from the pdfs and what isn't necessary from the pdfs.
    - Created a new plan to start with strictly images from the floor plans that were 100% necessary for the project.
        - Separated screenshots based on the use of them for the computer vision side of the project. This includes:
            - Schematics that show the basic layout, based on floor of the building
            - Elevation Drawings up to all 4 sides of the building
            - Schematics that show where the building is placed (elevation of ground, parking area, service line, stuff like that)
            - Schematics that show materials that make up the different elements in the basic layouts
            - Schematics that show different things that are placed inside the floors, (ex: Light Sources, Fans, Outlets, Furniture, etc)
            - Schematics that show electrical plans
            - Schematics that show things like ventilation.
            - Schematics that show roofing/rafters
            - Schematics that show additional elements like walls that are probably not needed
            - Images that show the key for each plan.
    - Tried and struggled to think of solutions that work for all 4 floor plans that the team received. The 4 plans are fairly significantly different from one another.
    - Considered the idea of having an AI generate floorplans with the ones the group has received and considered using additional floor plans found on the internet.
        - This has some issues that the group will need to ask the client about, specifically whether the group can even feed images into an AI.
            - Could possibly use Iowa State Gemini Pro if needed since it doesn't save chats, but Gemini would very likely be worse than Claude at this task.
- Jase Beaubien:
    - Did more research on graphics and planning.

Pending issues

- Nakshatra Gupta:
    - The extraction script worked on the first sample plan, but it did not work on the three other sample plans. It needs changes to handle plans that are formatted differently, which shows that plan layouts vary more than the first sample suggested.
- Rishiraj Parmar: N/A
- Ryan McNab:
    - The second floor is causing many issues with mesh collision, and the roof is not

being fully read correctly from the document.
    - The floors have incorrect scaling from each other, sometimes allowing the first floor to be smaller than the second.
- Daniel Hargrave:
    - Is the group required to care about electrical stuff, ventilation, specific wall design, etc.
    - Could the team feed images in Claude or other AI to create more plans for training?
    - Can the YOLOv11 model be used? The client is looking into themselves and told the team that they would get back to the group on this issue.
    - The current approach is just taking screenshots which is not the long term plan for this project. Need to figure out a plan to take only the stuff that is needed from the document, even when documents are inconsistent from one another.
        - No idea how this will work currently
- Jase Beaubien: N/A

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should be short.)*	**Hours this week**	**HOURS cumulative**
Nakshatra Gupta	Sample plan analysis, extraction script, website update, advisor and client communication.	4	16
Rishiraj Parmar	Developed an initial multi-story reconstruction integration prototype that organizes story-level data and validates floor relationships, elevations, heights, stairs, and open-to-below connections.	4	16
Ryan McNab	Used the new sample blueprints to improve the prototype, but left it in a somewhat broken state without enough iteration.	6	21
Daniel Hargrave	Studied the documents provided and tried to separate screenshots into different categories based on what each does.	4	17

	Attempted to come up with some basic plans for how to move forward given the new information.		
Jase Beaubien	Planning and research for graphics.	3	15

Comments and Extended Discussion

Current Buildertrend microsoft teams accounts do not work as intended, so communication is still getting stuck at the moment.

Plans for the upcoming week

- Nakshatra Gupta:
  - Find out why the script failed on the other three plans and update it to handle them and share the results with the team.
- Rishiraj Parmar:
  - Coordinate the prototype field names and story structure with the team's shared JSON schema.
- Ryan McNab:
  - Iterate on the sample documents to improve the mesh collision mainly. Will also improve roofing and scaling along the way, and may add an even larger catalog of materials.
- Daniel Hargrave:
  - Try and start working on making a model that could work with the screenshots that are available.
    - Will start with the basics for now. Won’t move into the other sections until progress is made with this.
- Jase Beaubien:
  - Further discuss and coordinate the object format for the deserialized plans.